USN | | | | | | | | | | | [24MTEE103]

FIRST SEMESTER M.Tech EXAMINATION, JANUARY - 2025

ADVANCED CONCRETE TECHNOLOGY

Time:3 Hours **Maximum Marks:100**

Instructions: **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 20 marks.

ii. Use of code books IS-10262-2009, ACI, DOE permitted

			CO's	BL	Marks
1.	**a.**	How do you determine the optimum dosage of superplasticizer using Marsh cone?	**1**	**4**	**06**
	b.	Define and list mineral admixtures. Explain the effect of flyash on hardened state of concrete.	**1**	**3**	**14**
2.	**a.**	List the tests conducted on hardened concrete. Explain the tension test on concrete specimen.	**2**	**2**	**10**
	b.	Why NDT is required in concrete test? Explain principle and procedure of the ultrasonic pulse velocity method.	**2**	**5**	**10**
3.	**a.**	Explain the factors affecting the mix design of concrete.	**3**	**2**	**06**
	b.	Design M60 grade high strength concrete mix as per IS:10262. The following are the design data's for proportioning:	**3**	**5**	**14**

Grade designation	: M60
Type of cement	: OPC53 grade confirming to IS12269
Type of Flyash	Class F & sp. Gravity is 2.2
Silica fumes	20000 m^2/kg (Surface area) & sp. Gravity is 2.2
Max. nominal size of agg.	: 20mm
Shape of CA	Angular
Workability	: 125mm (slump)
Exposure condition	: Severe (as defined in IS 456)
Min. cement content	: 400 kg/m^3
Chemical admixture (HRWRA)	Super plasticizer confirming to IS 9103
Specific gravity of HRWRA	1.1 , solid content = 42 %
Specific gravity of cement	: 3.15
Sieve analysis of FA	: Confirming to zone-I of IS 383

Material	**Coarse aggregate**	**Fine aggregate**
Specific gravity	2.8	2.7
Water absorption	0.4%	1.0%
Free surface moisture	nil	nil

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		OR			
4.	**a.**	Enumerate the need for SCC and mention its applications	**3**	**3**	**06**
	b.	Design a concrete mix in ACI method for 250 mm thick JPCP pavement. The desired properties are: • Slump = 25 mm • Average 28-day flexural strength of at least 4.5 MPa; after conversion factor assume compressive strength 51.7Mpa • Coarse aggregate: nominal maximum size = 37.5 mm; dry-rodded weight = 1600 kg/m^3, specific gravity = 2.68, moisture content = 1.0 percent, absorption = 0.5 percent • Fine aggregate: fineness modulus = 2.80, specific gravity = 2.64, moisture content = 5%, absorption = 0.7% • Air content: 4.5 – 6.5% • Maximum allowable water-cement ratio = 0.44 • Minimum cement content = 335 kg • Density of water = 1000 kg/m^3 • Specific gravity of portland cement = 3.15	**3**	**5**	**14**
5.	**a.**	Mention the methods controlling sulphate attack.	**4**	**3**	**10**
	b.	State the factors influencing the permeability of concrete. Explain how size of aggregate affect permeability?	**4**	**3**	**10**
		OR			
6.	**a.**	What is carbonation of concrete? How does it influence the corrosion of steel?	**4**	**4**	**08**
	b.	How to prevent the effect of freezing and thawing in concrete. Write any two tests for assessment of frost damage in concrete.	**4**	**3**	**12**
7.	**a.**	What are the different aggregates that would be used in light weight concrete? Mention the demerits of light weight concrete.	**5**	**2**	**06**
	b.	Design a light weight concrete mix to suit the following requirements, **i.** specified 28days compressive strength=12N/mm^2; **ii.** control factor=0.8; **iii.** Type of aggregate=leftag (FA) and leca (CA); **iv.** Required workability=High; **v.** Relative density (air)=1.3; Fine and coarse aggregate have 4% and 5% moisture content respectively. Use relevant codes/charts.	**5**	**5**	**14**
		OR			
8.	**a.**	Explain balling effect and orientation of fibers.	**5**	**3**	**08**
	b.	What are the different types of polymer concrete? Explain polymer modified concrete and its applications.	**5**	**2**	**12**